Lab 5

Joystick Control of an RC Servo

Ronald Nguyen

Nguyen.rona@northeastern.edu

Ayaan Gupta

Gupta.ayaa@northeastern.edu

Professor Thomas R. Consi

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Abstract

The purpose of this lab was to control an RC servo with an analog joystick using the ADC on the DE1-SoC board. The rc_servo circuit from Lab 4 was reused, with its switch input replaced by the joystick's X-axis reading. LPM multiplier and divider blocks converted the 12-bit ADC reading to an angle from 0° to 180°, which set the servo position and the 7-segment display. The pulse width and joystick voltage were measured on an ADALM 2000 oscilloscope at three positions, and the circuit was demonstrated to the TA.

Introduction

An analog joystick is two potentiometers at right angles, each wired as a voltage divider between 5 V and ground. The X-axis voltage is read by the 12-bit ADC on the DE1-SoC, giving a raw reading from 0 to 4095. For the prelab, the reading was scaled to an angle from 0° to 180°:

$$\theta = (180 \times \text{ADC}) / 4095,\ 0 \leq \text{ADC} \leq 4095$$

The multiplication is done first since dividing by 4095 first would only give 0 or 1. The angle is then converted to a pulse width with the Lab 4 equation, $N = 500\theta + 30{,}000$ clock periods, which gives 0.6 ms at 0° and 2.4 ms at 180°.

Setup

This lab was completed in Quartus Prime Lite with an Intel DE1-SoC board. A new project, joystick_servo, was created with the rc_servo files from Lab 4. The ADC controller was added from the IP Catalog for the DE1-SoC with a 12.5 MHz ADC clock and 8 channels, along with LPM_MULT, LPM_DIVIDE, and LPM_CONSTANT blocks. The joystick was powered from the ADC header, and its VRx pin shared a breadboard row with ADC_IN0 and scope channel 2. The servo, level shifter, and scope channel 1 were wired the same as in Lab 4. The new connections are listed in Table 1.

Table 1: Joystick and oscilloscope connections

Signal	FPGA Pin	Header Pin	Connected To
+5 V	—	ADC 1	Joystick +5V
Ground	—	ADC 10	Joystick GND
ADC_IN0	—	ADC 2	Breadboard row with VRx, CH2+
Ground	—	—	ADALM 2000 CH2− (breadboard)
GPIO1_35 (PWM)	PIN_AC22	JP2 40	Level shifter to servo
GPIO1_34 (Scope)	PIN_AA20	JP2 39	ADALM 2000 CH1
Ground	—	JP2 12	ADALM 2000 GND

Design

The circuit, shown in Figure 1, has three parts: the ADC interface, the angle conversion, and the rc_servo block from Lab 4.

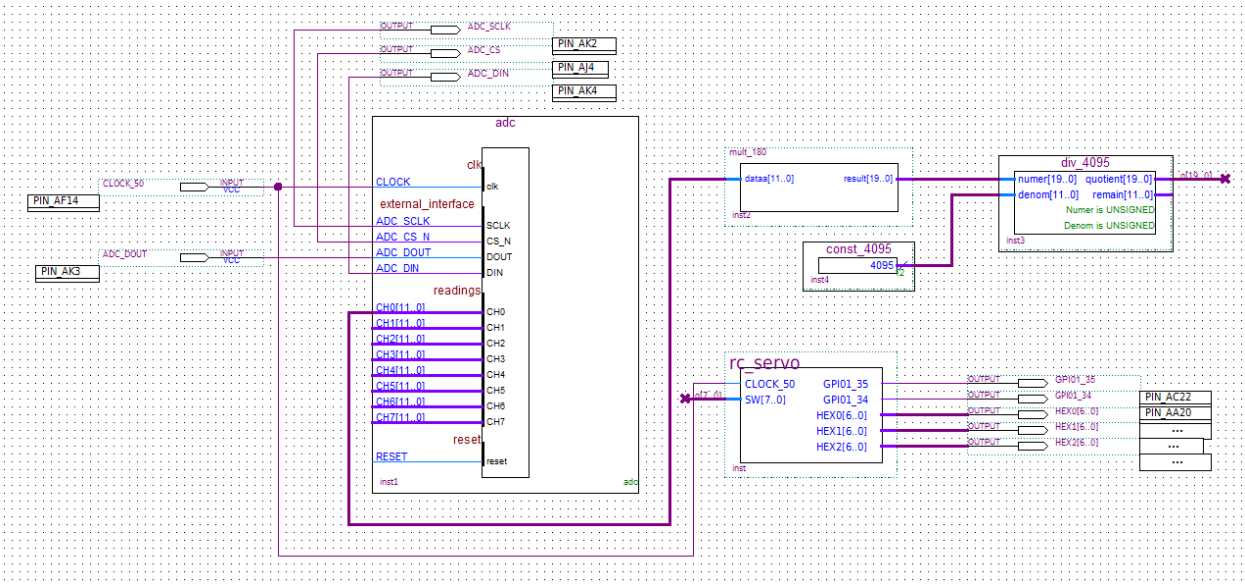


Figure 1: Schematic of the joystick servo circuit

Reading the Joystick with the ADC

The adc block is clocked by CLOCK_50 (PIN_AF14) and reads the ADC chip over a serial interface on ADC_SCLK (PIN_AK2), ADC_CS_N (PIN_AJ4), ADC_DIN (PIN_AK4), and ADC_DOUT (PIN_AK3). It converts all 8 channels continuously, but only CH0[11..0] is used. The other channels and reset are unconnected.

Converting the Reading to an Angle

Mult_180, an unsigned LPM_MULT, multiplies CH0 by 180. The largest product, 4095 × 180 = 737,100, fits in 20 bits. Div_4095, an unsigned LPM_DIVIDE, divides it by 4095 from const_4095, a 12-bit LPM_CONSTANT. The quotient is at most 180, so only its lower 8 bits, q[7..0], connect to rc_servo.

Driving the Servo

Rc_servo is the Lab 4 circuit made into a symbol, with the angle on SW[7..0] in place of the switches. It shows the angle on HEX0 through HEX2 and outputs a pulse of 500θ + 30,000 clock periods every 50 ms on GPIO1_35 and GPIO1_34. The HEX pins were assigned in reverse order as in Labs 3 and 4.

Results & Analysis

Debugging the Circuit

When first compiled, the 20-bit quotient could not connect directly to the 8-bit SW input, so the two were joined with named bus stubs, q[19..0] and q[7..0]. The GPIO pins also had to be renamed to match the imported Lab 4 assignments. The Programmer failed because the project was compiled for a 5CSEMA6 instead of the board's 5CSEMA5, which was fixed by changing the device and recompiling. Quartus then reported the ADC modules declared twice, which was fixed by keeping only adc.qip in the project, and the lost pin assignments were added back to the .qsf. On the board, the joystick had no effect until its ADC header wiring was corrected, and the display showed an upside-down A until the HEX pin assignments were reversed. Last, scope channel 2 read near 0 V at every position until its negative lead was grounded. The TA reviewed the circuit and approved the demonstration.

Measurements

Only one pulse fit on screen, so the pulse width was measured with the time cursors, and the joystick voltage was read from the channel 2 mean. The three positions are shown in Figures 2 through 4 and compared in Table 2. The expected width is 0.6 ms + 0.01 ms per displayed degree, and the expected angle is $\theta = 180 \times V / 4.096$, explained below.

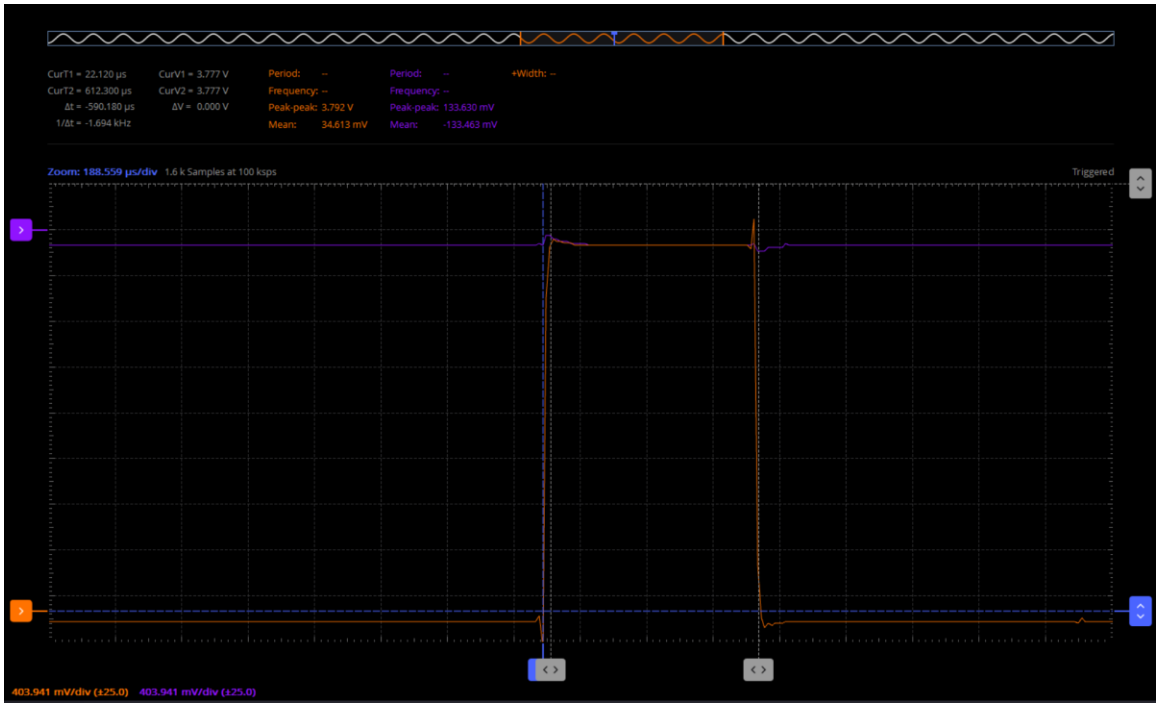


Figure 2: PWM signal at 0°, +Width = 0.590 ms, joystick voltage = −0.133 V

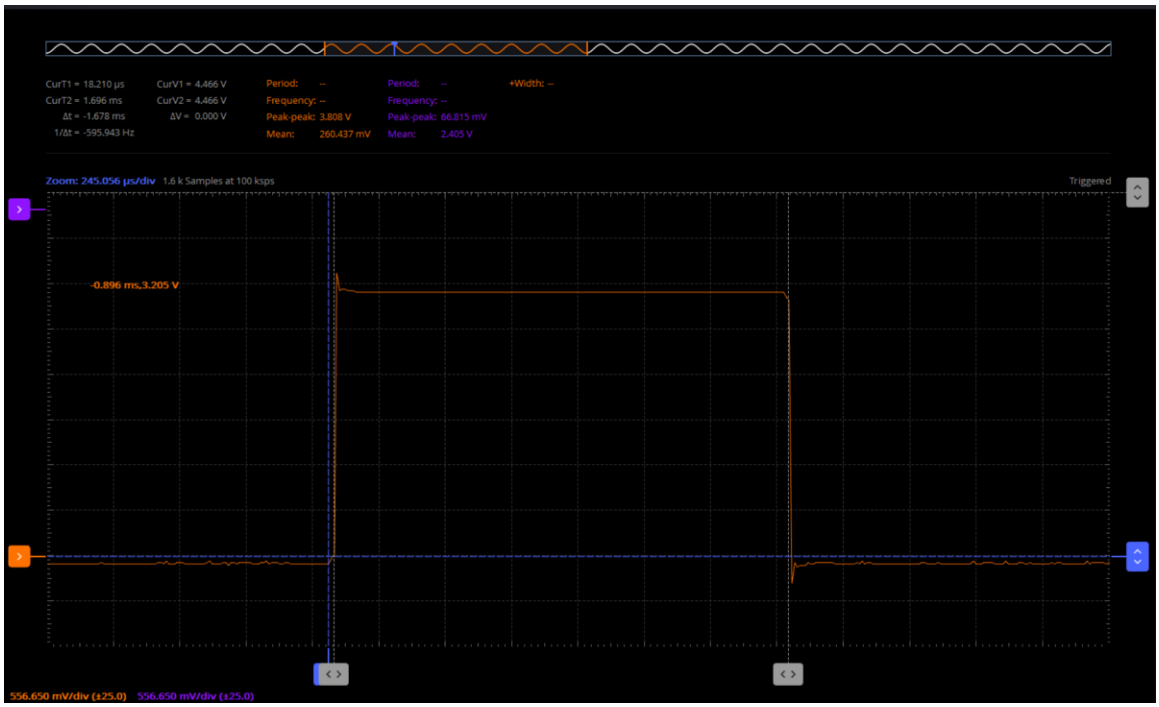


Figure 3: PWM signal at 108°, +Width = 1.678 ms, joystick voltage = 2.405 V

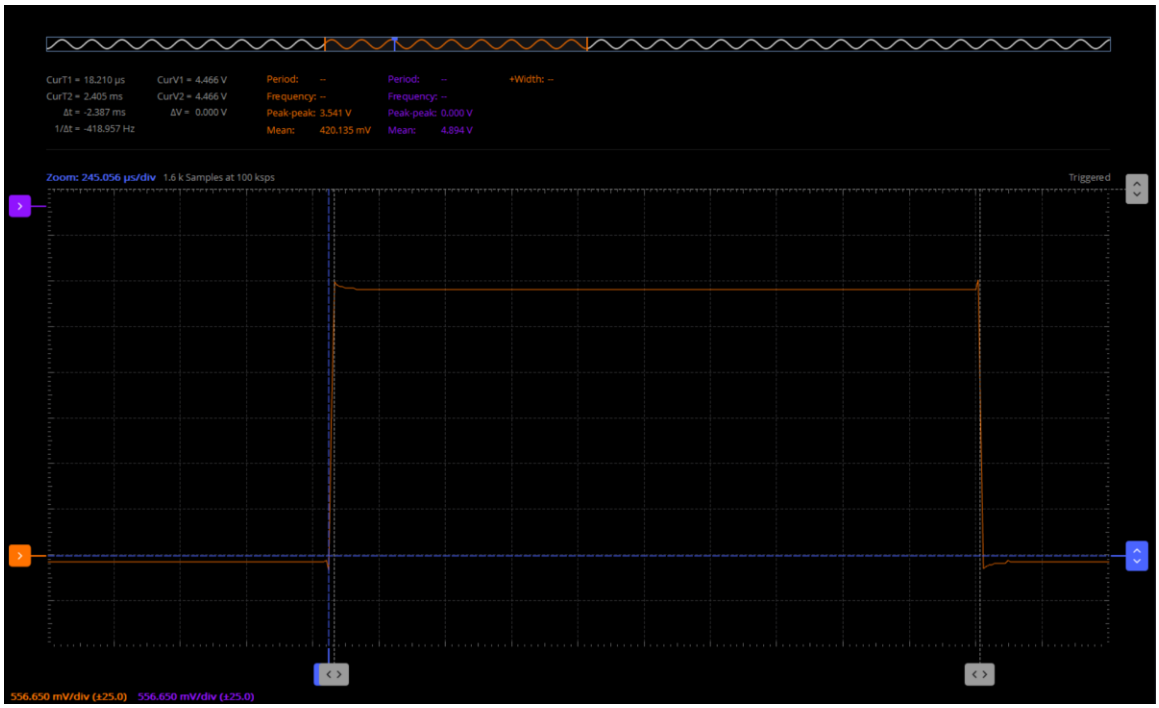


Figure 4: PWM signal at 180°, +Width = 2.387 ms, joystick voltage = 4.894 V

Table 2: Comparison of expected, measured, and displayed values

Joystick Voltage	Expected Angle (°)	Displayed Angle (°)	Expected Width	Measured Width	Error
−0.133 V	0	0	0.600 ms	0.590 ms	1.7%
2.405 V	105	108	1.680 ms	1.678 ms	0.1%
4.894 V	180	180	2.400 ms	2.387 ms	0.5%

All pulse widths were within 0.013 ms of the expected values, about one sample at 100 ksps, so the error came from placing the cursors by hand.

The angles did not follow the 0 to 5 V range given in the lab manual. At 2.405 V, a 5 V scale gives 86°, but the display showed 108°. The DE1-SoC's ADC chip, the LTC2308, has a 0 to 4.096 V input range, which gives 105°. The remaining 3° is about 0.07 V, within the scope's offset seen at 0°. Above 4.096 V the reading saturates at 4095, so the servo reaches 180° before the joystick is fully pushed, as seen at 4.894 V. Figure 5 shows the operating circuit.

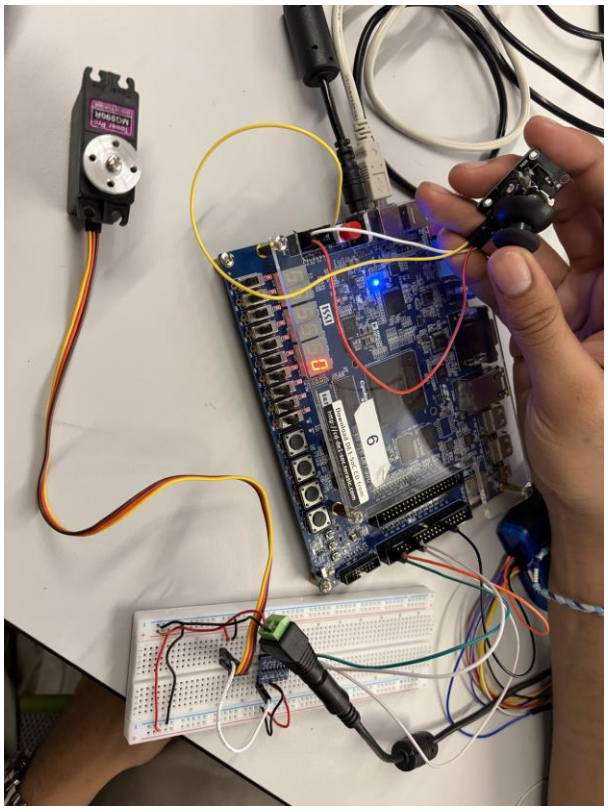


Figure 5: Operating circuit with the joystick, 7-segment displays, and RC servo at 0°

Conclusion

The joystick controlled the RC servo through the DE1-SoC's ADC, using $\theta = (180 \times ADC) / 4095$ and the rc_servo block from Lab 4. The pulse widths matched the expected values within the scope's resolution, and the displayed angles matched the joystick voltage once the ADC's 0 to 4.096 V range was taken into account.

References

[1] "Lab 5: Joystick Control of an RC Servo," Northeastern University, EECE 2160, Fall 2026.
[2] "Using the DE-Series ADC Controller," Intel FPGA University Program.
[3] "DE1-SoC User Manual," Terasic Technologies.